FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Main)-2020

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. Loss in KE = $k_i - k_f$

$$= \frac{1}{2}(40)\times 4^2 + \frac{1}{2}(60)(2)^2 - \frac{1}{2}(100)\times(2.8)^2 = 48 \text{ J.}$$

2. B

Sol. The charge q is completely closed by four identical given box. Therefore, the required flux = $\frac{1}{4}\frac{q}{\in_o}$.

3. A

Sol. $N = N_o e^{-\lambda t}$

$$0.99 = 1\, e^{-\lambda t} \Rightarrow \lambda t = \ln\left(\frac{100}{99}\right)$$

$$50 \text{ sec} = t = \frac{\ln\left(\frac{100}{99}\right)}{\lambda}$$

$$\Rightarrow \quad t_{1/2} = \frac{50\ln(2)}{\ln(100/99)}$$

4. A

Sol. $v_I = -m^2 v_o$

$$= -\frac{9}{25}\times(-5\hat{i}) = 1.8\,\hat{i}$$

5. B

Sol. $\frac{1}{r}=\frac{1}{r_1}-\frac{1}{r_2} \Rightarrow$ r = 4 mm.

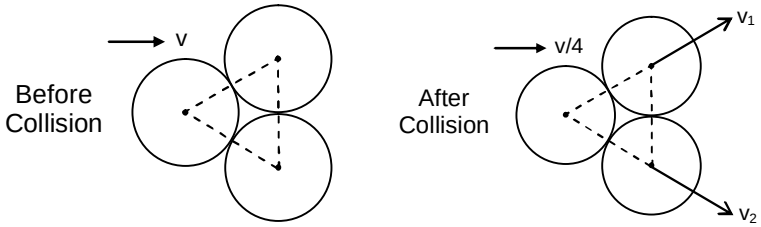
6. D

Sol. $\vec{\tau}=\vec{\mu}\times\vec{B}$

$= (i\vec{A})\times\vec{B}$

7. C

Sol.



$2mv_1\cos 30° + m\frac{v}{4} = mv$ …(i)

$e = \frac{v_1-\frac{v}{4}\cos 30^o}{v\cos 30^o}$ …(ii)

8. C

Sol. $P=\rho(g+a)h$

9. B

Sol. At steady state.

Rate of generation of A = Rate of decay of A.

$q = \lambda N_A$

$\Rightarrow N_A=\frac{q}{\lambda}$

10. B

Sol. $x = 2\sin 3t$

$y = 2-2\cos 3t$

$\sin^2 3t+\cos^2 3t = 1$

11. A

Sol. $f\lambda = v = \frac{10\lambda v}{d} \Rightarrow \lambda=\frac{d}{10}$.

Maximum path diff. can be $\frac{d}{3}$. Hence, number of maximum will be 6.

12. B

Sol. Displacement method

$\frac{u}{D-u}=2$ (I) & D – u – u = 30 cm ...(II)

$3u = 2D$ (I) & $D - 2u = 30$ cm

And $\frac{1}{f} = \frac{1}{D-u} + \frac{1}{u}$

$\Rightarrow$ f = 20 cm.

13. A

Sol. Inside water

u = –20 cm

f = –15 cm

$\Rightarrow$ v = –60 cm

$m = \frac{-v}{u}$ = –3

$\Rightarrow$ n = 3

14. D

Sol. $\tau = \frac{i_{finel}}{\left(\frac{di}{dt}\right)_{initial}} = \frac{\left(\frac{2}{10}\right)}{\left(\frac{\Delta V_L}{L}\right)}$

$= \frac{\frac{1}{5}}{\frac{2}{3}\times 2} = \frac{3}{20}$ seconds.

15. C

Sol. The net resistance across the cell is decreased to R/2 hence the current delivered by the battery will increase while potential across R will remain same.

16. D

Sol. $\phi_B = 0$

$\because \vec{B} \perp \vec{S}$

$\therefore \frac{d\phi_B}{dt} = 0$

17. A

Sol. As λ increases saturation current also increases.

18. C

Sol. $\Delta V = \left|\frac{d\phi}{dt}\right| = (\text{Area}) \times \frac{dB}{dt} = \frac{\pi R}{2}\left(\frac{R}{2}\right)(B_o)$

$\Delta V_{AB} = \frac{\pi R^2}{4} B_o$

19. A

Sol. Applying momentum conservation in the direction of string

$mu\frac{\sqrt{3}}{2} = 2\,mv \Rightarrow v = \frac{\sqrt{3}}{4}(u)$

20. C

Sol. $\vec{N} + \vec{F}_s + m\vec{g} = 0$

$\therefore \quad \left|\vec{N} + \vec{F}_s\right| = \left|-m\vec{g}\right| = 20$ N

SECTION – B

21. 6

Sol. $h = \dfrac{2T\cos\theta}{r\rho g}$

22. 8

Sol. Constraint equation.

SECTION – C

23. 00000.60

Sol. $\dfrac{2}{3}MR_1^2 = \dfrac{2}{5}MR_2^2$

24. 00000.20

Sol. I_L and I_C will be in opposite phases.

$I_{net} = I_L - I_C$

$= 0.6 - 0.4 = 0.2$ A

25. 00011.00

Sol. $I_1 = \left(\dfrac{s}{s+99}\right) \times I$

Chemistry

PART – II

SECTION – A

26. C

Sol. $P = K_2SO_4$, $Q = BaSO_4$, $R = BaS$, $S = H_2S$

27. B

Sol. $\Delta C_p = 2\times 25.1 + 3\times 75.3 - (103.8 + 3\times 28.8) = 85.9 \text{ J/mole}$

$$\frac{\Delta H_{358} - \Delta H_{298}}{T_2 - T_1} = \Delta C_p$$

$\Delta H_{358} = -28.136 \text{ kJ/mole}$

28. B

Sol. $HIn \rightleftharpoons H^+ + In^-$

$$K_a = \frac{\left[H^+\right]\left[In^-\right]}{\left[HIn\right]} = \frac{\left[H^+\right]\left[base\right]}{\left[Acid\right]}$$

$$\text{or, } \left[H^+\right] = Ka\frac{\left[Acid\right]}{\left[Base\right]}$$

For 75% red, $\left[H^+\right] = \frac{Ka\times 75}{25} = \frac{3\times 10^{-5}\times 75}{25} = 9\times 10^{-5}$; $pH = 4.05$

For 75% blue, $\left[H^+\right] = \frac{\left(3\times 10^{-5}\right)\times 25}{75} = 1\times 10^{-5}$; $pH = 5$

The change in pH = 5-4.05 = 0.95

29. C

Sol. $t_{1/2}$ is independent of concentration of reactant for 1st order reaction

30. A

Sol. (A) As we know that,
Resonance energy = $\Delta H°$ (observed) – $\Delta H°$ (calculated)
= (50.4 – 232.6) kJ / mol = – 182.2 kJ mol^{-1}

31. B

Sol. In case of weak acid heat of neutralization is greater than – 13.7 KCl / mole because extra energy is required to dissociate the acid.

32. D

Sol. Sulphur forms two sigma bonds with two lone pairs.

33. B

Sol. $\frac{\Lambda_m}{\Lambda_e} = 3 \Rightarrow \Lambda_m = 3\Lambda_e$

n-factor of $MgCl_2.KCl.6\ H_2O = 3$.

34. C

Sol. HCOOH is stronger than CH_3COOH. Hence the solution of HCOOH will contain more no. of ions than that of CH_3COOH.

35. A

Sol. Aliphatic amines are stronger base than aromatic amines

36. B

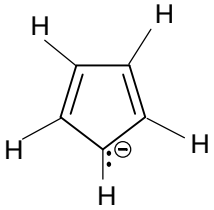
Sol. In (B) the NH_2 group which is at a larger distance from the COOH group, should be more basic it should accept H^+ ion from acid.

37. B

Sol. One Na^+ is removed and remaining three(effective no. per unit cell) are present.

38. C

Sol. In cyclopentadienyl anion, each carbon is sp^2 hybridized

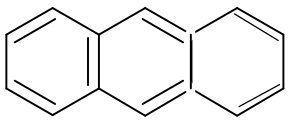


39. D

Sol. $E^0_{cell} = E^0_{cell} > 0$ (due to 1 M concentration of ions)

40. D

Sol.



41. B

Sol. Halogen atoms except fluorine form compounds with valency 1, 3, 5 and 7.

42. C

Sol. The fraction is $\frac{2}{3}$

43. C

Sol. Increases in pressure or addition of C affect the second equilibrium hence concentration of A_2 changes which further affect the first equilibrium. Addition of inert gas at constant volume does not affect the equilibrium

44. A

Sol. $Na_2B_4O_7.10H_2O \xrightarrow[-10H_2O]{\text{heat}} Na_2B_4O_7 \xrightarrow{740^oC} \underset{\text{Sodium metaborate}}{2NaBO_2} + B_2O_3$

$$B_2O_3 + CuSO_4 \longrightarrow CuO.B_2O_3 + SO_3$$
$$\downarrow$$
$$Cu(BO_2)_2 \text{ (blue)}$$
Copper metaborate

45. D

Sol. At normal boiling point (500 K), the following equilibrium exists between the two physical states at 1 atm.

$$X(\text{liquid}) \rightleftharpoons X(\text{gas})$$

For the equilibrium $\Delta G = 0$

If the pressure increases the change shifts to backward direction for which ΔG = +ve.

If pressure decreases the change shifts towards forward direction, for which ΔG = -ve.

If temperature increase, the change will proceed towards forward direction, for which ΔG = -ve.

SECTION – B

46. 3

Sol. $5 \rightarrow 2, 4 \rightarrow 2, 3 \rightarrow 2$

47. 4

Sol. $A(g) + 2B(g) \rightleftharpoons 2C(g) + D(g)$

a	1.5 a	0	0
(a - x)	$1.5^a - 2x$	2x	2x

Now, $a - x = x$

$\Rightarrow x = a/2$

SECTION – C

48. 00028.66

Sol. $\frac{1}{\lambda} = 1.09 \times 10^7 \times 36\left[\frac{1}{1} - \frac{1}{9}\right]$

49. 00001.50

Sol. SF_4 $\ell p = 1$

SF_6 $\ell p = 0$

H_2O $\ell p = 2$

50. 00002.50

Sol. Na^+, Mg^{2+}, O^{2-}, F^- & Cl^- will have more radius

So, $\frac{x}{2} = 2.5$

Mathematics

PART – III

SECTION – A

51. B

Sol. $f'(x)-\frac{2x(x+1)}{x+1}f(x)=\frac{e^{x^2}}{(x+1)^2}$

$IF = e^{\int -2x\,dx} = e^{-x^2}$

$\Rightarrow \quad f(x).\left(-e^{-x^2}\right)=\int \frac{dx}{(x+1)^2}$

$\Rightarrow \quad f(x).e^{-x^2}=\frac{1}{x+1}+C$

At x = 0, f(0) = 5

$\Rightarrow \quad C = 6$

$\therefore \quad f(x)=\left(\frac{6x+5}{x+1}\right).e^{x^2}$

Hence, (b) is the correct answer.

52. D

Sol. Clearly, R is reflexive and symmetric.

Let z_1Rz_2 and z_2Rz_3

$\Rightarrow \quad \frac{z_1-z_2}{z_1+z_2}$ is real, say k

and $\frac{z_2-z_3}{z_2+z_3}$ is real, say l

Now, $\frac{z_1-z_2}{z_1+z_2}=k \Rightarrow \frac{z_1}{z_2}=\frac{1+k}{1-k}$

and $\frac{z_2-z_3}{z_2+z_3}=l \Rightarrow \frac{z_2}{z_3}=\frac{1+l}{1-l}$

Thus, $\frac{z_1}{z_3}=\frac{(1+k)(1+l)}{(1-k)(1-l)}$

$\Rightarrow \quad \frac{z_1-z_3}{z_1+z_3}=\frac{(1+k)(1+l)-(1-k)(1-l)}{(1+k)(1+l)+(1-k)(1-l)}$

which is real.

53. A

Sol. Since A.M. $\geq$ G.M., We get

$$\frac{1}{2}\left[\sqrt{x^2+x}+\frac{\tan^2\alpha}{\sqrt{x^2+x}}\right]\geq\sqrt{\sqrt{x^2+x}\times\frac{\tan^2 x}{\sqrt{x^2+x}}}=|\tan\alpha|$$

54. B

Sol. All the function in (a), (b), (c) are defined on $[-1, 1]$. Let

$g(x) = -x^2 + x + \sin x + \cos x - \log(1+|x|)$. (The choice of g is clear as $x^2, \cos x \log(1+|x|)$ are even function and $x, \sin x$ are odd function). Then

$g(-x) = -x^2 - x - \sin x + \cos x - \log(1+|x|) = -(x^2 + x + \sin x - \cos x + \log(1+|x|)) = -f(x)$.

55. C

Sol. According to the given conditions, $c_1^2 + c_2^2 + c_3^2 = 1$, $a.c - 0$, $b.c = 0$ and

$$\frac{\sqrt{3}}{2} = \cos\frac{\pi}{6} = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}\sqrt{b_1^2 + b_2^2 + b_3^2}}$$

Thus $a_1c_1 + a_2c_2 + a_3c_3 = 0$, $b_1c_1 + b_2c_2 + b_3c_3 = 0$ and

$$\frac{\sqrt{3}}{2}\left(a_1^2 + a_2^2 + a_3^2\right)^{1/2}\left(b_1^2 + b_2^2 + b_3^2\right)^{1/2} = a_1b_1 + a_2b_2 + a_3b_3$$

Now, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}^2 = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

$$= \begin{vmatrix} a_1^2 + a_2^2 + a_3^2 & a_1b_1 + a_2b_2 + a_3b_3 & a_1c_1 + a_2c_2 + a_3c_3 \\ a_1b_1 + a_2b_2 + a_3b_3 & b_1^2 + b_2^2 + b_3^2 & b_1c_1 + b_2c_2 + b_3c_3 \\ a_1c_1 + a_2c_2 + a_3c_3 & b_1c_1 + b_2c_2 + b_3c_3 & c_1^2 + c_2^2 + c_2^2 \end{vmatrix}$$

$$= \begin{vmatrix} a_1^2 + a_2^2 + a_3^2 & a_1b_1 + a_2b_2 + a_3b_3 & 0 \\ a_1b_1 + a_2b_2 + a_3b_3 & b_1^2 + b_2^2 + b_3^2 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right) - \left(a_1b_1 + a_2b_2 + a_3b_3\right)^2$$

$$= \left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right) - \frac{3}{4}\left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)$$

$$= \frac{1}{4}\left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)$$

56 D

Sol. Put $\frac{1}{2}\cos x = t$, so that $-\sin x\,dx = 2dt$ and $I = \int_{1/2}^{-1/2} e^{|t|}(2\sin t + 3\cos t)(-2)dt$

As $e^{|t|}\sin t$ is an odd function, and $e^{|t|}\cos t$ is an even function,

$$I = 12\int_0^{1/2} e^t \cos t\, dt = 12e^t \cos t\Big]_0^{1/2} + 12\int_0^{1/2} e^t \sin t\, dt$$

$$I = 12\left[\sqrt{e}\cos\left(\frac{1}{2}\right) - 1\right] + 12e^t \sin t\Big]_0^{1/2} - 12\int_0^{1/2} e^t \cos t\, dt$$

$$\Rightarrow 2I = 12\sqrt{e} = \left(\cos\left(\frac{1}{2}\right) + \sin\left(\frac{1}{2}\right) - 1\right).$$

57. A

Sol. Let $t = x + \sqrt{x^2 + b^2}$

$\Rightarrow \frac{1}{t} = \frac{1}{x + \sqrt{x^2 + b^2}} = \frac{\sqrt{x^2 + b^2} - x}{b^2}$

$\therefore\ t - \frac{b^2}{t} = 2x$ and $t + \frac{b^2}{t} = 2\sqrt{x^2 + b^2}$

Thus $2(a - x)\left(x + \sqrt{x^2 + b^2}\right) = \left(2a - t + \frac{b^2}{t}\right)(t)$

$= 2at - t^2 + b^2 = a^2 + b^2 - \left(a^2 - 2at + t^2\right)$

$= a^2 + b^2 - (a - t)^2$

Therefore, $y = 2(a - x)\left(x + \sqrt{x^2 + b^2}\right) \le a^2 + b^2$.

Hence, the maximum value of y is $a^2 + b^2$. This value is attained at $t = a$ or $x = \frac{\left(a^2 - b^2\right)}{2a}$.

58. C

Sol. $\tan^{-1}\frac{1}{1 + n(n+1)} = \tan^{-1}\frac{n + 1 - n}{1 + n(n+1)}$

$= \tan^{-1}(n+1) - \tan^{-1}(n)$

So that L.H.S. of the given equation is

$\tan^{-1}2 - \tan^{-1}1 + \tan^{-1}3 - \tan^{-1}2 + ... + \tan^{-1}(n+1) - \tan^{-1}n$.

$= \tan^{-1}(n+1) - \tan^{-1}1 = \tan^{-1}\frac{n + 1 - 1}{1 + (n+1)}$

$= \tan^{-1}\frac{n}{n+2}$

So that $\tan^{-1}\frac{n}{n+2} = \tan^{-1}\theta = \theta = \frac{n}{n+2}$.

59. C

Sol. Let $x_0 = \cos\theta$, then $x_1 = \sqrt{\frac{1}{2}(1 + \cos\theta)} = \cos\frac{\theta}{2}$, $x_2 = \cos\left(\frac{\theta}{2^2}\right)$, $x_3 = \cos\left(\frac{\theta}{2^3}\right)$.... and so on.

So that $\left[\frac{\sqrt{1 - x_0^2}}{x_1 x_2 x_3 .....\text{to infinite}}\right]$

$= \frac{\sin\theta}{\cos\frac{\theta}{2}\cos\frac{\theta}{2^2}.....\cos\frac{\theta}{2^n}.....\text{infinite}}$

$= \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\cos\frac{\theta}{2}\cos\frac{\theta}{2^2}.....\cos\frac{\theta}{2^n}.....\text{infinite}}$

$$= \frac{2^2 \sin\frac{\theta}{2^2}\cos\frac{\theta}{2^2}}{\cos\frac{\theta}{2^2}\cos\frac{\theta}{2^3}.....\cos\frac{\theta}{2^n}.....\text{infinite}}$$

$$= \lim_{n\to\infty} \frac{2^n \sin\frac{\theta}{2^n}}{\cos\frac{\theta}{2^{n+1}}} = \lim_{n\to\infty} \theta\left(\frac{\sin\frac{\theta}{2^n}}{\frac{\theta}{2^n}}\right)\frac{1}{\cos\frac{\theta}{2^{n+1}}} = \theta$$

So that $\cos\left[\frac{\sqrt{1-x_0^2}}{x_1 x_2.....\text{inf}}\right] = \cos\theta = x_0$

60. D

Sol. We have $f(x) = x^2 + 2bx + 2c^2 = (x+b)^2 + 2c^2 - b^2$

$\Rightarrow$ $\min f(x) = 2c^2 - b^2$

Also $g(x) = -x^2 - 2cx + b^2$

$= b^2 + c^2 - (x+c)^2$

so $\max g(x) = b^2 + c^2$

Since $\min f(x) > \max g(x)$,

so $2c^2 - b^2 > b^2 + c^2$

$\Rightarrow$ $c^2 > 2b^2 = |c| > \sqrt{2}|b|$.

61. D

Sol. $|\vec{a}| = 1 = |\vec{b}|$

$$\phi(\theta) = \int_{-\cos^2\theta}^{\sin^2\theta} f^2(x).dx$$

$$2f(x) = \frac{2x}{x^2} \Rightarrow \boxed{\frac{1}{x} = f(x)}$$

$$\phi(\theta) = \int_{-\cos^2\theta}^{\sin^2\theta} \frac{1}{x^2}.dx$$

$$= \left[\frac{-1}{x}\right]_{-\cos^2\theta}^{\sin^2\theta}$$

$$= \left(-\text{cosec}^2\theta\right) - \left(\sec^2\theta\right)$$

$$= -\text{cosec}^2\theta.\sec^2\theta$$

$$\phi(\theta) = \frac{-4}{\sin^2 2\theta} = -4\,\text{cosec}^2 2\theta$$

Fundamental period is $\left(\frac{\pi}{2}\right)$

62. D

Sol. $x = -10$ is the directrix of the given parabola. Hence the circle S = 0 will touch it. P_1P_2 is the diameter. So $\angle P_1P\,P_2 = 90^o$ and hence ΔP_1PP_2 is right angled and its orthocentre is point P.
Tangents at extremeties of focal chord are perpendicular. Hence in their combined equation (coefficient x^2) + (coefficient y^2) = 0.

63. C

Sol. 5^{5^5} is an odd natural number.

$\therefore x = 5^{5^5} = 5^{2m+1} = \left(25^m\right).5$

$m \in N$

$x = \left(24+1\right)^m.5 = 5 + a$ multiple of 24

Hence Remainder = 5

64. B

Sol. $f\left(g\left(x\right)\right) = f\left(\left|3x+4\right|\right)$

$-5 \le 3x+4 \le 5$

$x \in \left[-3, \dfrac{1}{3}\right]$

65. D

Sol. $y = \left(\sin^{-1}x\right)^4 + \left(\cos^{-1}x\right)^4$

$$\frac{dy}{dx} = \frac{4\left(\sin^{-1}x\right)^3}{\sqrt{1-x^2}} - \frac{4\left(\cos^{-1}x\right)^3}{\sqrt{1-x^2}} = 0$$

$\Rightarrow \sin^{-1}x = \cos^{-1}x$

$\Rightarrow x = \dfrac{1}{\sqrt{2}}$

$f\left(1\right) = \dfrac{\pi^4}{16}$

$f\left(-1\right) = \dfrac{\pi^4}{16} + \pi^4 = \dfrac{17\pi^4}{16}$ (Maximum)

$F\left(\dfrac{1}{\sqrt{2}}\right) = \dfrac{\pi^4}{128}$ (Minimum)

66. D

Sol. Let $f\left(x\right) = 3\tan x + x^3 - 2$

Then $f'\left(x\right) = 3\sec^2 x + 3x^2 > 0$

Hence $f\left(x\right)$ increases.

Also $f\left(0\right) = -2$ and $f\left(\dfrac{\pi}{4}\right) > 0$

So, by intermediate value theorem, $f\left(c\right) = 2$ for some c in $\left(0, \dfrac{\pi}{4}\right)$

Hence $f\left(x\right) = 0$ has only one root.

67. B

Sol. $\vec{b}.\vec{c}=0$

$\tan\alpha=-2,3$

$\tan\alpha=-2$

$\Rightarrow\alpha\rightarrow 2^{nd}$ or 4th quadrant

$\tan\alpha=3$

$\Rightarrow\alpha\rightarrow 1^{st}$ or 3rd quadrant

Now,

$\vec{a},\hat{k}<0$

$\sin 2\alpha<0$

$(2n-1)\pi<2\alpha<2n\pi$

$\frac{(2n-1)\pi}{2}<\alpha<n\pi$

$\alpha\rightarrow 2^{nd}$ quadrant

or, $\tan\alpha=-2$

or, $\alpha=(2n+1)\pi-\tan^{-1}2$

68. C

Sol. $y=u^m$

$\frac{dy}{dx}=m.u^{m-1}\frac{du}{dx}$

The given differential equation becomes $2x^4u^m.\,mu^{m-1}\frac{du}{dx}+u^{4m}=4x^6$

or $\frac{du}{dx}=\frac{4x^6-u^{4m}}{2mx^4x^{2m-1}}$

For homogeneous equation, degree should be same for numerator and denominator,

hence $6=4m=4+2m-1\Rightarrow m=\frac{3}{2}$

69. A

Sol. $(x^2-7x+6)(x^2-7x+12)+10$

Taking first and last terms together and middle terms together. Now make quadratic in $x^2-7x=t$.

We get, $(t+6)(t+12)+10$

$=t^2+18t+82$

$D=-4<0$

Hence value is always positive $\forall x\in R$.

70. C

Sol. Second diagonal will pass through the centre of the hyperbola.

SECTION – B

71. 1

Sol. In the 1st case,

P (they match) $= \frac{{}^{3}C_{2} + {}^{n}C_{2}}{{}^{n+3}C_{2}} = \frac{1}{2}; \frac{6 + n(n-1)}{(n+3)(n+2)} = \frac{1}{2}$

$\Rightarrow 2(n^2 - n + 6) = n^2 + 5n + 6$

$\Rightarrow n^2 - 7n + 6 = 0$

$\Rightarrow n = 1$ or 6

In the 2nd case,

$\frac{3}{n+3} \times \frac{3}{n+3} + \frac{n}{n+3} \times \frac{n}{n+3} = \frac{5}{8}$

Solving, $n^2 - 10n + 9 = 0$

n = 9 or 1 ……(ii)

From equation (i) and (ii), n = 1

72. 9

Sol. Given,

$a + 19d = \log_{10} 20$ ……(1)

$a + 31d = \log_{10} 32$ ……(2)

(2) – (1)

$12d = \log_{10} \frac{32}{20} = \log_{10} 16 - 1$

$12d = 4\log_{10} 2 - 1$

$\log_{10} 2 = \frac{12d + 1}{4}$ ……(A)

Again (2) + (1)

$2a + 50d = \log_{10} 640 = 6\log_{10} 2 + 1$

$\log_{10} 2 = \frac{2a + 50d - 1}{6}$ ……(B)

$\therefore \frac{12d + 1}{4} = \frac{2a + 50d - 1}{6} \Rightarrow 36d + 3 = 4a + 100d - 2$

$4a + 64d = 5$

$\underbrace{a + 16d}_{17^{th}\text{ term}} = \frac{5}{4}$

Hence, 17th term is rational and its value is $\frac{5}{4} = \frac{p}{q} \Rightarrow (p + q) = 9$

SECTION – C

73. 00000.25

Sol. Given $\int_0^1 f(x)dx = \frac{1}{3} + \int_0^1 \left(f\left(x^2\right)\right)^2 dx$

Consider, $I_1 = \int_0^1 \left(f\left(x^2\right)\right)^2 dx$

Put $x^2 = t \Rightarrow 2x\,dx = dt$

$I_1 = \frac{1}{2}\int_0^1 \frac{(f(t))^2}{\sqrt{t}}\,dt$

$\therefore$ Equation (1) becomes

$\int_0^1 \frac{t}{2\sqrt{t}}\,dt + \int_0^1 \frac{(f(t))^2 - 2\sqrt{t}\,f(t)}{2\sqrt{t}}\,dt \Rightarrow \int_0^1 \left(f(t) - \sqrt{t}\right)^2 dt = 0$

$\therefore f(t) = \sqrt{t}$

So, $f\left(\frac{1}{4}\right) = \frac{1}{2}$

So, reciprocal of $f\left(\frac{1}{4}\right) = 2$

74. 00000.12

Sol. Given curve is ellipse having centre (0, 0). All chords of ellipse through origin are bisected at origin

$\Rightarrow p = q = 0$

75. 00001.87

Sol. Given, $z + \omega = i$ ……(1)

and $z^2 + \omega^2 = 1$ …….(2)

$\therefore$ From (1), on squaring we get

$z^2 + \omega^2 + 2z\omega = -1 \Rightarrow 1 + 2z\omega = -1$ [Using (2)]

$\Rightarrow zw = -1$ …..(3)

Now, let us consider a quadratic equation in x whose roots are z and ω.

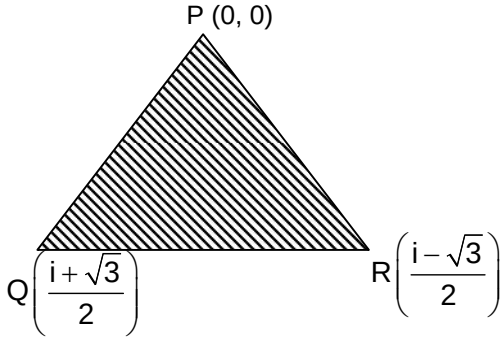
$\Rightarrow x^2 - ix - 1 = 0$

$\therefore x = \frac{i \pm \sqrt{i^2 + 4}}{2} = \frac{i \pm \sqrt{3}}{2}$

Let



$\omega = \frac{i + \sqrt{3}}{2}$ $\qquad$ $z = \frac{i - \sqrt{3}}{2}$



So, ar.$(\Delta PQR) = \frac{1}{2}\begin{Vmatrix} 0 & 0 & 1 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 1 \\ \frac{-\sqrt{3}}{2} & \frac{1}{2} & 1 \end{Vmatrix} = \frac{1}{2}\left(\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4}\right) = \frac{\sqrt{3}}{4}$